ROYAL UNIVERSITY OF PHNOM PENH

Faculty of Engineering

Data Vault 2.0 Implementation for Cambodia Tax System Data Warehouse Using Medallion Architecture

A Final Report — Master of Science in Data Science and Engineering

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Background

GDT Digital Transformation

Cambodia's General Department of Taxation (GDT) is modernizing its digital systems to improve tax collection, compliance monitoring, and revenue forecasting. With thousands of taxpayers filing monthly and annual declarations, transactional data volumes continue to grow rapidly.

Growing Tax Data Volumes

Financial institutions, government agencies, and regulatory bodies increasingly rely on historical tax data for economic policy, compliance enforcement, and resource allocation. This demands an enterprise-grade data warehouse capable of handling large-scale analytical workloads.

Kimball Star Schema Limits

The Kimball dimensional model struggles when source systems change frequently. Star schema requires significant restructuring when new tax types or reporting requirements are introduced, causing costly downtime and redevelopment effort in government IT environments.

Inmon 3NF Limitations

The Inmon corporate information factory produces complex ETL pipelines and slow analytical queries. Its normalized Enterprise Data Warehouse approach lacks the agility needed for modern data integration, parallel development, and evolving regulatory compliance requirements.

Data Vault 2.0 offers agility, auditability, and scalability — the ideal solution

Problem Statement

CRITICAL

Schema Rigidity

Star schema models require significant restructuring when source systems change, which happens frequently as government regulations evolve. When a new tax type is introduced or reporting requirements change, the entire dimensional model may need redesign, causing costly downtime.

CRITICAL

Historical Tracking

Traditional approaches struggle to maintain complete historical records. Regulatory bodies require full audit trails showing what data was recorded, when it changed, and previous values. SCD Type 1 overwrites history entirely, while SCD Type 2 requires upfront design decisions.

HIGH

Scalable ETL

Growing tax data volumes demand scalable ETL processes handling both full and incremental loads efficiently. Many implementations rely on manual processes lacking error handling, logging, and recovery mechanisms. Failed loads have no systematic way to identify or restart.

HIGH

Business Rules Gap

There is a gap between raw operational data and business-ready analytical information. Tax administrators need derived metrics such as compliance scores, payment behavior patterns, and revenue trends requiring complex business logic applied consistently across the data warehouse.

These challenges motivate the Data Vault 2.0 + Medallion Architecture approach

Challenges: The Four V

V	Dimension	GDT Challenge	Architectural Response
V1	**Volume**	GDT retains all taxpayer records for legal audit with no deletion. Volume grows continuously and never shrinks.	Watermark-based incremental ETL processes only changed records per cycle. Insert-only Bronze keeps query time constant.
V2	**Velocity**	Monthly deadlines create sharp ingestion spikes with thousands of submissions in days. Daily payments add secondary bursts.	ETL_Watermark delta capture extracts only modified records. 49-step pipeline with Propagate=False error isolation.
V3	**Variety**	Three structure data categories: lookup, reference, and transaction across 9 tables. Future integration with customs, e-invoicing, banking.	DV2.0 Hub-Satellite-Link separates keys from attributes. New sources integrate via Hubs with zero restructuring.
V4	**Veracity**	Incorrect compliance scores trigger unwarranted audits with legal consequences. Bronze errors propagate silently through Silver and Gold.	12-point Source DB Verification with zero violations across 58,813 records. HashDiff + ETL_StepLog ensure data integrity.

Aim & Objectives

Research Aim: Design & implement a complete DV2.0 + Medallion data warehouse for Cambodia tax administration

O1 — Medallion Architecture

Design a 9-table source database simulating Cambodia's tax system and implement a 4-layer medallion architecture (Staging, Bronze, Silver, Gold) using Data Vault 2.0 Hub-Satellite-Link modeling across 6 databases on SQL Server 2025.

O3 — POC Validation

Validate through 4 proof-of-concept demos: Schema Rigidity (zero-impact satellite addition), Historical Tracking (insert-only audit trail), Scalable ETL (step-level error recovery), and Business Rules (compliance scoring via Business Vault).

O2 — ETL Control Framework

Develop 60 SSIS packages (3 Master, 8 Orchestrator, 49 Child) supporting full and incremental loading with watermark-based change detection. Implement batch-and-step logging, step-level error recovery, and SCD2 history tracking across all layers.

O4 — Gold Layer & Implementation Guide

Create business vault constructs (PIT, Bridge, Business tables) and produce a gold-layer star schema (7 Dims, 4 Facts) for BI dashboards. Provide a Technical Implementation Guide with full SSIS package inventory for reproducibility.

Research Gap

Theoretical Focus

Most Data Vault 2.0 studies remain theoretical or conceptual. Yessad and Labiod present a comparative study of Inmon, Kimball, and Data Vault but provide no complete end-to-end implementation or reproducible proof-of-concept evaluation with measurable outcomes.

Government Tax Domain Gap

No published end-to-end DV2 + Medallion Architecture implementation exists for government tax administration in academic literature. Existing work focuses on telecommunications or healthcare domains where regulatory compliance needs differ significantly.

No End-to-End Framework

Existing studies focus on individual aspects — modeling patterns, ETL performance, or schema evolution — rather than presenting a complete end-to-end implementation spanning all four medallion layers (Staging, Bronze, Silver, Gold) with integrated ETL control frameworks.

Limited ETL Governance

Few studies address integrated ETL control frameworks combining batch logging, step tracking, watermark-based incremental loading, and error recovery within a Data Vault architecture. Production-grade pipeline observability remains largely unaddressed in academic research.

Gap: No published DV2 + Medallion implementation for government tax systems

Contributions

Complete 4-Layer Implementation

A complete, reproducible four-layer medallion architecture (Staging, Bronze, Silver, Gold) using Data Vault 2.0 on SQL Server 2025, comprising 67 database objects across six databases and 60 SSIS packages. This is the first published end-to-end DV2 + Medallion implementation for government tax administration.

Integrated ETL Control Framework

An integrated ETL control framework with batch logging, step tracking, watermark-based incremental loading, and error handling providing production-grade pipeline observability. The framework demonstrates step-level recovery and achieves effective change detection with SAT versioning and SCD2 history tracking.

Implementation Guide & DB Verification

A comprehensive Technical Implementation Guide as a companion reproducibility artifact detailing the complete SSIS package inventory, deployment procedures, and source database verification with data quality validation framework for full research reproducibility.

POC Evaluation

A proof-of-concept evaluation demonstrating how Data Vault 2.0 addresses four key challenges in Cambodia tax data warehousing: schema flexibility, historical tracking, scalable ETL error recovery, and business rules derivation. Each POC validated with measurable before/after SQL evidence.

First end-to-end DV2 + Medallion implementation for Cambodia tax administration

Literature Review

Year	Study	Key Contribution	Limitation / Gap	Our Approach
Data Vault 2.0 & Medallion Architecture				
2015	Linstedt	Hub/Link/Satellite modeling; hash keys; insert-only audit trail	Foundation methodology; no layered architecture pattern	**DV2 + 4-layer Medallion (Staging→Bronze→Silver→Gold)**
2021	Armbrust et al.	Medallion (Bronze/Silver/Gold) for lakehouse pattern	Cloud-native focus; no DV2 integration	**DV2 structures mapped to each Medallion layer**
ETL Frameworks & Data Quality				
2002	Vassiliadis et al.	ETL conceptual modeling; pipeline optimization	No DV2 context; no error recovery	**Control framework + watermark + error isolation**
Related Work — Data Vault Implementations				
2016	Yessad & Labiod	3-way comparison (Inmon/Kimball/DV)	No implementation; purely theoretical	**Complete impl. with POC evidence**
2016	Naamane & Jovanovic	Empirical study with telecom data	No medallion arch.; no Gold layer	**4-layer medallion + Gold star schema**
2024	Vines	TPC-DS benchmark; performance focus	No domain-specific; no ETL control	**Cambodia tax domain + ETL framework**
2025	Singh	Resilience & adaptability analysis	Theoretical analysis; no implementation	**Full POC with measurable evidence**
2026	**This Study**	**DV2 + Medallion + ETL + POC evaluation**	**Simulated data (verified, 12-check)**	**First gov. tax DV2 end-to-end impl.**

Source Database & Sample Data

TaxSystemDB — 9 Source Tables (3 Lookup + 3 Reference + 3 Transaction)

Table	Type	Description	Records
Category	Lookup	Taxpayer categories (Individual, SME, Corp)	**5**
Structure	Lookup	Business structure types (Individual, Partnership, Corp)	**5**
Activity	Lookup	Business activity sectors (Manufacturing, Service, etc.)	**5**
Taxpayer	Reference	Registered taxpayers with TaxID, business name	**1,000**
Officer	Reference	Tax collection officers with department	**10**
Owner	Reference	Business owners with ownership percentages	**1,956**
MonthlyDeclaration	Transaction	Monthly tax declarations with revenue, tax, penalties	**38,491**
Payment	Transaction	Tax payments linked to declarations	**13,733**
AnnualDeclaration	Transaction	Annual tax declarations with yearly summaries	**2,785**

Data Generator (02_TransactionData.sql): 1,000 taxpayers, 4 years (2020–2023), ~80% monthly filing rate, 58,813 records. Verified: 10 FKs, 19 CHECK constraints, 35 indexes, 0 orphans, 0 NULLs (see Source Database Verification document).

Analytical Questions

■ Strategic Analytics

01 Revenue Trends

What are the monthly and annual tax revenue patterns by taxpayer category, business structure, and activity type across 2020–2023?

Gold FACT_MonthlyDeclaration

02 Compliance Scoring

How can taxpayer compliance be measured using declaration timeliness, payment rates, and penalty frequency?

Silver BUS_ComplianceScore

03 Payment Behavior

What are the payment completion rates and outstanding balances by taxpayer segment and declaration period?

Gold FACT_Payment

04 Risk Assessment

Which taxpayers show declining compliance or late declaration trends that require enforcement action?

Gold FACT_DeclarationLifecycle

These questions drive the transformation: Source OLTP → Staging → Bronze (DV2.0) → Silver (Business Vault) → Gold (Star Schema)

System Architecture

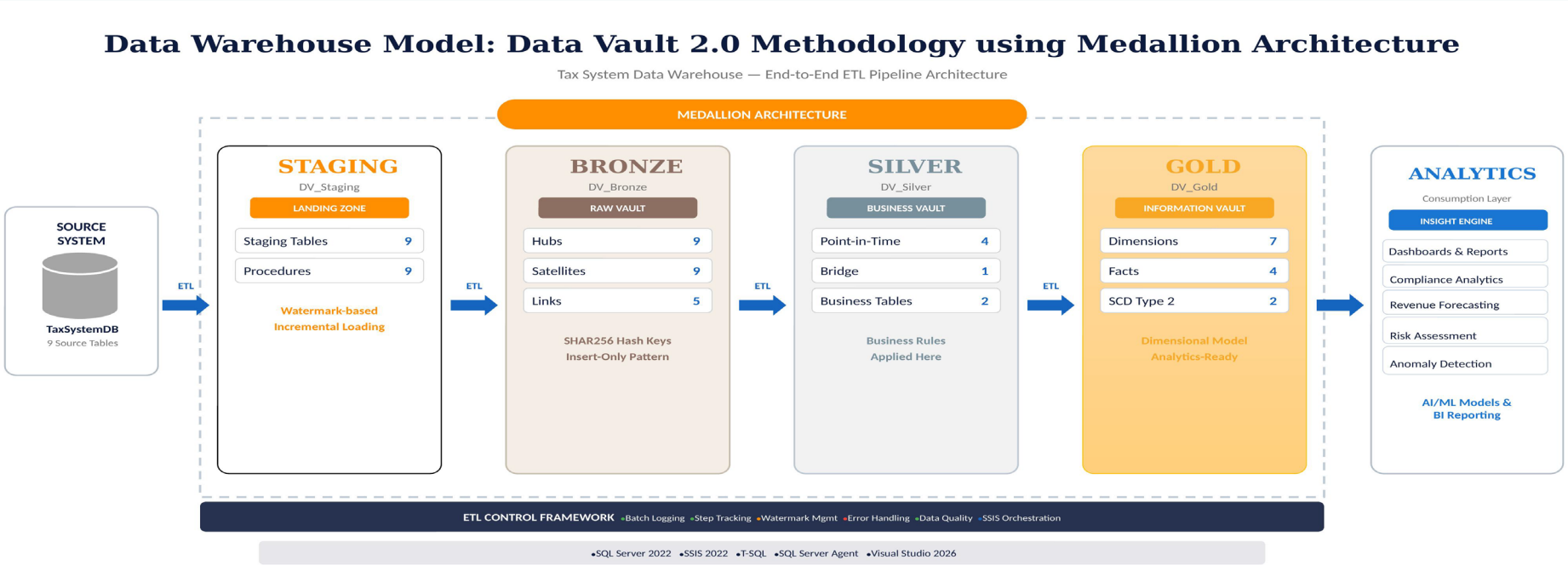

Figure 3.1: Data Warehouse Model: Data Vault 2.0 Methodology using Medallion Architecture

Technology Stack

Software Specification

Technology	Version	Purpose
SQL Server 2025	Developer Edition (RTM 17.0)	Database engine, SSIS runtime, SQL Server Agent
SSIS	Included with SQL Server 2025+	ETL package execution and orchestration
T-SQL	—	SSIS packages, Data Flow tasks, ETL logic
SQL Server Agent	Included with SQL Server 2025	Job scheduling for automated pipeline execution
Visual Studio Community	2026	SSIS package design and development (IDE)

Deployment Environment

Component	Specification	Details
Cloud Platform	Google Cloud Compute Engine	e2-standard-4 instance
CPU	4 vCPUs (2 cores)	Intel Broadwell or newer
Memory	16 GB RAM	12 GB allocated to SQL Server
Storage	128 GB SSD	Persistent disk, single volume
Operating System	Windows Server 2025 Datacenter	Server with Desktop Experience

Implementation — Four Layers

Staging (9)

- Full: TRUNCATE → INSERT
- Incr: Watermark delta
- **1:1 source mapping**

Bronze (23)

- 9 Hubs: SHA-256 hash keys
- 9 SATs: HashDiff + SCD2
- 5 Links: Relationships
- **Insert-only audit trail**

Silver (6)

- 3 PIT: Temporal snapshots
- 1 Bridge: Pre-joined
- **2 BUS: Compliance scores**

Gold (11)

- 7 DIMs: SCD1 + SCD2
- 4 FACTs: Monthly, Payment
- Snapshot, Audit
- **Star schema for BI**

Master Orchestration — 49 Steps

Init Batch → Staging (9) → Hubs (9) → SATs (9) → Links (5) → Silver (6) → Gold (11) → Close Batch

Analytical Questions → Implementation

How each analytical question is answered by the DV2 + Medallion architecture (Gold star schema from DV_Gold database)

01 Revenue Trends
Monthly/annual tax revenue patterns by taxpayer category, structure, and activity (2020–2023)
→ **Gold FACT_MonthlyDeclaration**
Star schema fact joins DIM_Taxpayer, DIM_Category, DIM_Structure, DIM_Activity for BI reporting

02 Compliance Scoring
Measure compliance via declaration timeliness, payment rates, and penalty frequency
→ **Silver BUS_ComplianceScore**
Business Vault derives compliance ratings from declaration timeliness, payment behavior, penalty history

03 Payment Behavior
Payment completion rates and outstanding balances by taxpayer segment and period
→ **Gold FACT_Payment**
Fact table with payment amounts, method, status joined to DIM_Taxpayer and DIM_PaymentMethod

04 Risk Assessment
Identify taxpayers with declining compliance or late declaration trends for enforcement
→ **Gold FACT_DeclarationLifecycle**
Tracks declaration submission, status changes, late filing patterns with DIM_Taxpayer and DIM_Status

Data Flow Through Layers

Source OLTP
TaxSystemDB
9 tables | 58K records
→
Staging
DV_Staging
9 tables
→
Bronze (DV2.0)
DV_Bronze
9 Hubs | 9 SATs | 5 Links
→
Silver (BV)
DV_Silver
3 PIT | 1 Bridge | 2 BUS
→
Gold (Star)
DV_Gold
7 DIM | 4 FACT

67 database objects across 6 databases • 60 SSIS packages • Full/Incremental load with watermark-based change detection

POC Evidence — 4 Challenges Addressed

C1: Schema Rigidity	Problem: ALTER TABLE impacts multiple dependent objects	DV2: CREATE new Satellite only — zero impact on 60 SSIS packages
C2: Historical Tracking	Problem: Limited audit trail, no full version history	DV2: Insert-only SATs preserve ALL versions
C3: Scalable ETL	Problem: Monolithic pipelines, no step-level logging or error recovery	DV2: ETL control with BatchLog/StepLog/ErrorLog, OnError Event Handlers
C4: Business Rules Gap	Problem: No separation of raw data from derived business logic	DV2: Silver applies business rules (ComplianceScore); Bronze preserves raw

Conclusion

This research successfully designed, implemented, and evaluated a complete Data Vault 2.0 data warehouse for the Cambodia tax system using medallion architecture on Microsoft SQL Server.

- Complete 4-layer architecture: 67 objects, 60 SSIS packages, 49 ETL steps across 6 databases
- ETL pipeline: Full and incremental loading with watermark-based change detection and per-step error isolation
- Gold layer star schema provides consumption-ready dimensional model for BI reporting
- Performance benchmarking and scalability analysis identified as future work
- POC validation: All 4 challenges (schema rigidity, history, scalable ETL, business rules) addressed
- 3 companion documents [32][33][34] provide full reproducibility with SQL scripts and SSMS screenshots

Future Work & Recommendations

1. Real-time processing with CDC (Change Data Capture) for critical compliance monitoring tables

2. Production deployment with real GDT data — data quality validation, security classification, access controls

3. Performance benchmarking: ETL load times, query performance (Gold vs Raw Vault), scalability analysis with columnstore indexes

4. BI dashboard integration (Power BI, Metabase) consuming Gold layer star schema for tax analytics

5. Multi-source integration: customs, banking, e-invoicing via Same-As Links & Multi-Active Satellites

6. Advanced analytics: ML models on Gold layer for compliance risk scoring, revenue forecasting, fraud detection, and anomaly detection

Thank You

Questions & Discussion

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